Antiderivatives and Indefinite Integrals

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic antiderivatives

Find each indefinite integral:

a $\int (3x^2 - 4x + 5)\,dx$

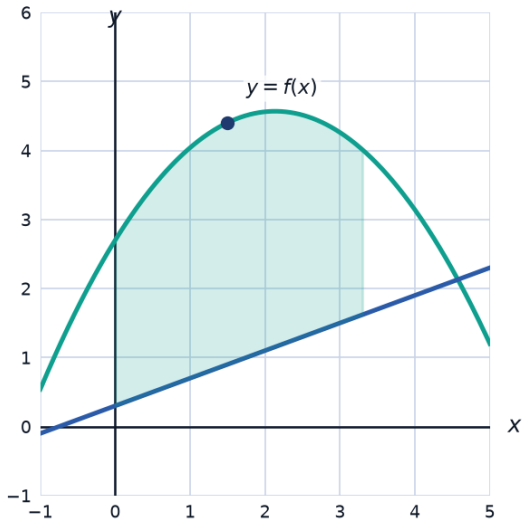
b $\int \frac{1}{x^2}\,dx$

c $\int \sqrt{x}\,dx$

d $\int \sec^2 x\,dx$

2 [Foundational] • Target: Using initial conditions

A particle moves along a line with velocity $v(t) = 3t^2 - 6t$ m/s. If its position at $t = 0$ is $s(0) = 4$ m, find its position function $s(t)$.



3 [Foundational] • Target: Using initial conditions

An object is dropped from rest, so its acceleration is $a(t) = -9.8$ m/s^2.

a Find the velocity function $v(t)$.

b If the object starts at height $s(0) = 45$ m, find its position function $s(t)$.

c Find how long it takes to hit the ground ($s = 0$), to two decimal places.

4 [Foundational] • Target: More antiderivatives

Find the general antiderivative of $f(x) = (x + 1)^3$ (expand first, or use the power rule on the whole expression, and state which method you used).

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5 [Foundational] • Target: Motion applications

A rocket's acceleration is $a(t) = 6t$ m/s^2, with $v(0) = 0$ and $s(0) = 0$.

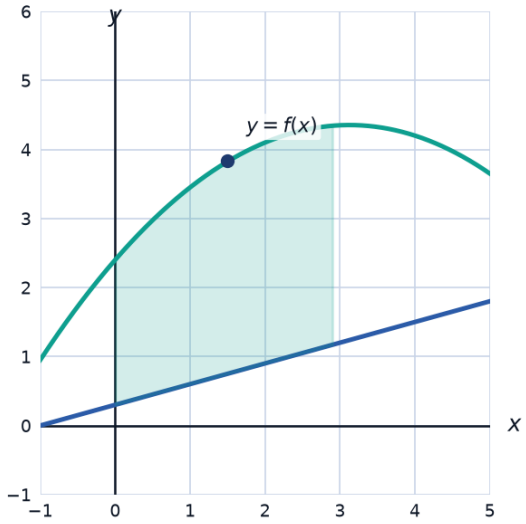
a Find $v(t)$.

b Find $s(t)$.

c Find the distance traveled in the first 4 seconds.

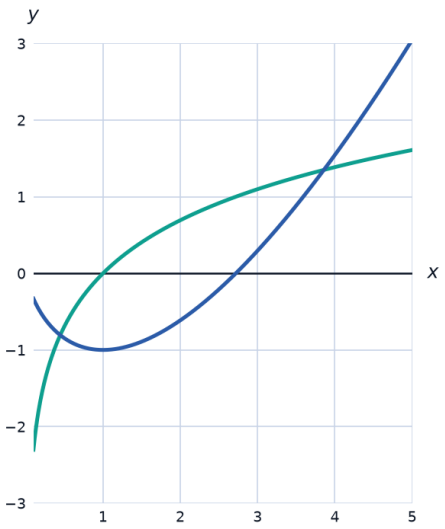
6 [Foundational] • Target: Antiderivatives of composite expressions

Find $\int (2x - 1)^5 \, dx$, treating $(2x - 1)$ as a linear inner function.



7 [Foundational] • Target: Verifying antiderivatives by differentiation

Verify that $F(x) = x\ln x - x$ is an antiderivative of $f(x) = \ln x$ by differentiating $F(x)$.



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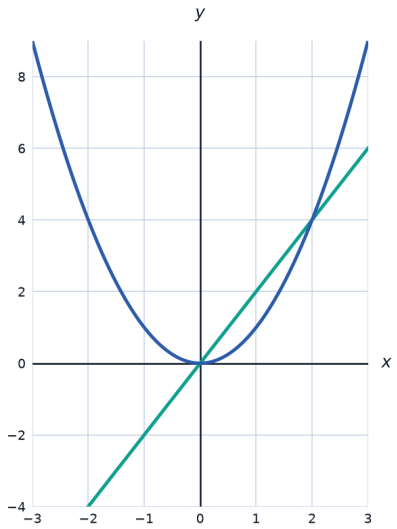
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: A projectile motion application**

A ball is thrown upward with initial velocity 15 m/s from a height of 2 m, so its acceleration is $a(t) = -9.8$ m/s^2. Find its velocity function $v(t)$ and position function $s(t)$.

9 **[Intermediate]** • **Target: Visualizing an antiderivative family**

The graph shows $f(x) = 2x$ and one member of its antiderivative family, $F(x) = x^2$. Every other antiderivative of f is a vertical shift of F.

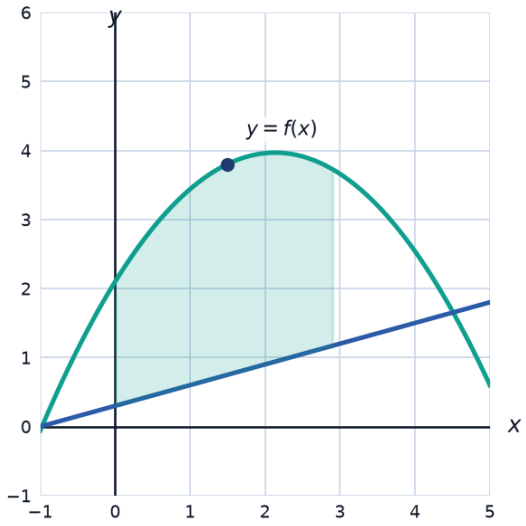


a Write the general antiderivative of $f(x) = 2x$.

b If a specific antiderivative passes through $(1, 4)$, find its constant C.

10 **[Intermediate]** • **Target: Antiderivatives requiring algebraic rewriting first**

Find $\int \frac{x^3 + 2x}{x}\,dx$ by simplifying the integrand before integrating.



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11 [Intermediate] • Target: Antiderivatives requiring algebraic rewriting first

Find $\int (x+1)(x-3)\,dx$ by expanding the product first.

12 [Intermediate] • Target: Using initial conditions

Level 2 · MediumSolve the transcendental equation analytically for x in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

13 [Intermediate] • Target: More antiderivatives

Level 2 · Medium Solve the initial value problem $\frac{dy}{dx} - y = e^{3x}$ with $y(0) = 2$.

14 [Intermediate] • Target: Motion applications

Level 1 · Foundations Find the area of the region bounded by $y = x^2$ and $y = 2x - x^2$.

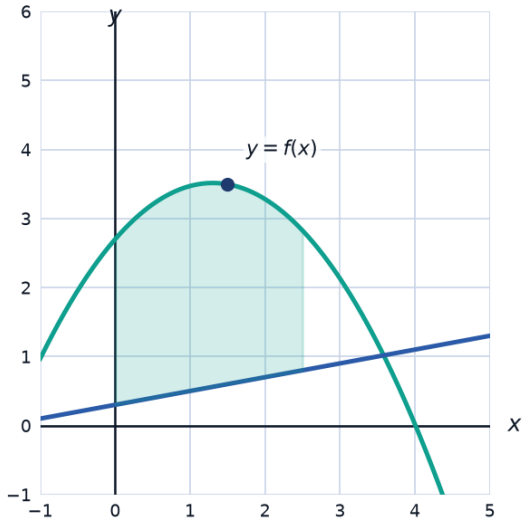
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Antiderivatives of composite expressions • Cross-Lesson: 12.02 Power Series and Taylor Series

Level 1 · Foundations Evaluate the indefinite integral $\int \frac{\cos(\ln x)}{x}\,dx$ for $x > 0$.

16 [Challenging] • Target: Verifying antiderivatives by differentiation • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

[Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



17 [Challenging] • Target: Antiderivatives of reciprocal and rational-power functions • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

[Applications of Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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18 **[Challenging]** • **Target: A projectile motion application** • **Cross-Lesson: 15.01 Double Integrals**

[Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 **[Challenging]** • **Target: Visualizing an antiderivative family** • **Cross-Lesson: 16.01 Vector Fields and Line Integrals**

[Multiple Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 **[Challenging]** • **Target: Antiderivatives requiring algebraic rewriting first** • **Cross-Lesson: 2.01 Evaluating Limits**

Level 3 · AdvancedLet $f(x) = x^5 + 3x^3 + 2x - 1$. Find the equation of the tangent line to the graph of $y = f^{-1}(x)$ at $x = 5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 **[Foundational]** • **Target: Basic antiderivatives**

The Padé approximant $[m/n]$ for $f(x)$ is a rational function $P_m(x)/Q_n(x)$ matching the Taylor series to order $m + n$. It typically converges more:

a. slowly than Taylor

b. only at 0

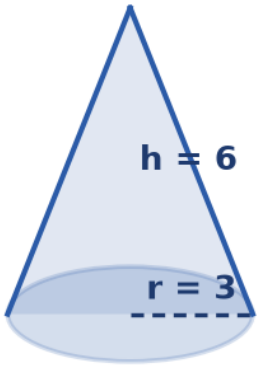
c. rapidly than Taylor for poles

d. never

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22 [Foundational] • Target: Using initial conditions

Water drains from a cone-shaped tank (radius 3, height 6) at $\frac{dV}{dt} = 2$ m^3/min. Using $V = \frac{1}{3}\pi r^2 h$ with $r = \frac{3}{6}h$, find $\frac{dh}{dt}$ when $h = 2$.



a. $\frac{2}{\pi}$

b. $\frac{1}{\pi}$

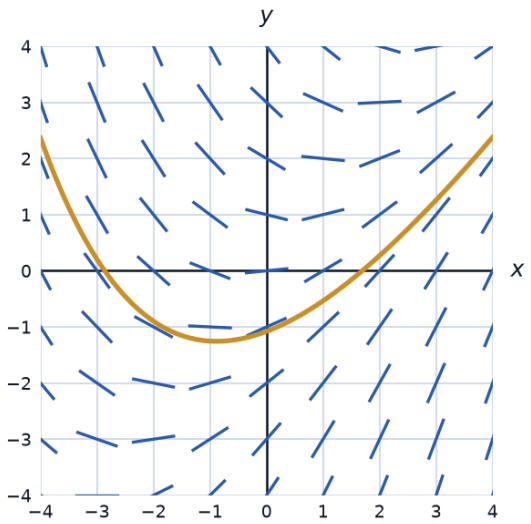
c. $-\frac{2}{\pi}$

d. $\frac{4}{\pi}$

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23 [Foundational] • Target: More antiderivatives

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.



a. $e^{-1} + 1$

b. $2e^{-1} + 1$

c. $e^{-1} - 1$

d. $2 - e^{-1}$

24 [Foundational] • Target: Motion applications

$\int \frac{1}{x^2 - 1}\, dx$ using partial fractions:

a. $\frac{1}{2}\ln|x - 1| - \frac{1}{2}\ln|x + 1| + C$

b. $\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$

c. $-\frac{1}{2}\ln|x^2 - 1| + C$

d. A and B are equivalent

25 [Foundational] • Target: Antiderivatives of composite expressions

Volume of cone: radius r, height h, using disc method:

a. $\frac{2}{3}\pi r^2 h$

b. $\frac{1}{2}\pi r^2 h$

c. $\pi r^2 h$

d. $\frac{1}{3}\pi r^2 h$

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26 [Foundational] • Target: Verifying antiderivatives by differentiation

Generalised binomial: $\sqrt{1-x} = \sum_{n=0}^{\infty} \binom{1/2}{n}(-1)^n x^n$. First three terms:

a. $1 - x - x^2$

b. $1 + x/2 - x^2/8$

c. $1 - x/2 - x^2/8$

d. $1 - x/2$

27 [Foundational] • Target: Antiderivatives of reciprocal and rational-power functions

The antiderivative of $f(x) = 0$ is:

a. 0

b. x

c. any constant C

d. undefined

28 [Foundational] • Target: A projectile motion application

Area of triangle with vertices at origin, $(3, 0)$, $(0, 4)$ using integration:

a. 3

b. 12

c. 6

d. π

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Visualizing an antiderivative family

Evaluate $\iint_{\mathbb{R}^2} e^{-\pi(x^2+y^2)} dA$: using Gaussian integral result $\int_{-\infty}^{\infty} e^{-\pi x^2} dx = 1$:

a. π

b. 1

c. π^2

d. $1/\pi$

30 [Intermediate] • Target: Antiderivatives requiring algebraic rewriting first

Volume of revolution using shells: $y = \cos(x^2)$, $x \in [0, \sqrt{\pi/2}]$, rotated about y-axis:

a. 2π

b. $\pi/2$

c. π

d. $\sqrt{\pi}$

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31 [Intermediate] • Target: Basic antiderivatives

Surface area of sphere radius R using surface area of revolution:

a. $4\pi R^2$

b. $3\pi R^2$

c. $2\pi R^2$

d. πR^3

32 [Intermediate] • Target: Using initial conditions

Power series integration: $\int_0^x \frac{1}{1+t^2}dt = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} =$

a. $\arctan x$

b. $\ln(1+x)$

c. $\sin x$

d. $x/(1+x^2)$

33 [Intermediate] • Target: More antiderivatives

Using linearisation to approximate $\sqrt{4.02}$:

a. 2.005

b. 2.01

c. 2.001

d. 2.002

34 [Intermediate] • Target: Motion applications

Volume of a sphere radius R using disc method:

a. $\frac{4}{3}\pi R^3$

b. $\frac{4}{3}\pi R^2$

c. $4\pi R^2$

d. $2\pi R^3$

35 [Intermediate] • Target: Antiderivatives of composite expressions

Power series represent real analytic functions. Is $f(x) = e^{-1/x^2}$ ($xe0$), $f(0) = 0$ analytic at 0?

a. No — all derivatives are 0 but $fe0$ near 0

b. Yes

c. Yes, since it is smooth

d. No, it is not smooth

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36 [Intermediate] • Target: Verifying antiderivatives by differentiation

Shooting angle for max range of a projectile (no air resistance):

a. 30°

b. 45°

c. 60°

d. 90°

37 [Intermediate] • Target: Antiderivatives of reciprocal and rational-power functions

If $f(0) = 1$, $f'(x) = f(x)^2$, and f has a power series, what is $f(x)$?

a. e^x

b. $\frac{1}{1-x}$

c. $\ln(1+x)$

d. $\cos x$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A projectile motion application • Cross-Lesson: 15.01 Double Integrals

Maclaurin series for e^x: first four terms:

a. $1 + x + x^2 + x^3$

b. $1 + x + x^2/2! + x^3/3!$

c. $x + x^2 + x^3$

d. $1 - x + x^2 - x^3$

39 [Challenging] • Target: Visualizing an antiderivative family • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Power series of $\arcsin x$: first two non-trivial terms beyond x:

a. $x + x^3/3$

b. $x - x^3/6$

c. $x + x^3/6 + \cdots$

d. $\pi/2 - x$

40 [Challenging] • Target: Antiderivatives requiring algebraic rewriting first • Cross-Lesson: 2.01 Evaluating Limits

The Gaussian integral: $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$. Using the polar trick $I^2 = \iint e^{-(x^2+y^2)} dA = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r \, dr \, d\theta$:

a. $I = \pi$

b. $I = \sqrt{\pi}$

c. $I = \pi/2$

d. $I = 1$

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41 [Challenging] • Target: Basic antiderivatives • Cross-Lesson: 2.02 Continuity

Equation of motion in rotating frame: $m\mathbf{a}_{\text{rot}} = \mathbf{F} - 2m\boldsymbol{\omega} \times \mathbf{v} - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$. The term $-2m\boldsymbol{\omega} \times \mathbf{v}$ is the:

a. centrifugal force

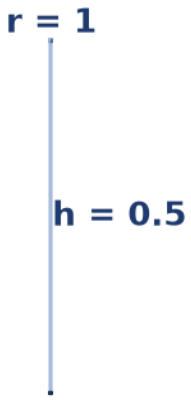
b. Coriolis force

c. gravity

d. normal force

42 [Challenging] • Target: Using initial conditions • Cross-Lesson: 3.01 The Derivative from First Principles

A water tank (cylinder, radius 1 m) drains. If height decreases at 0.5 m/min, water volume leaving per minute:



a. 0.5π

b. π

c. 2π

d. 4π

43 [Challenging] • Target: More antiderivatives • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Which of the following series converges?

a. $\sum_{n=2}^{\infty} \frac{1}{2n+1}$

b. $\sum_{n=2}^{\infty} \frac{1}{n \ln(n)^2}$

c. $\sum_{n=2}^{\infty} \left(1 + \frac{1}{n}\right)^n$

d. $\sum_{n=2}^{\infty} \frac{n}{n^2+1}$

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44 [Challenging] • Target: Motion applications • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Which of the following series diverges?

a. $\sum_{n=2}^{\infty} \left(\frac{n}{2n+1}\right)^n$

b. $\sum_{n=2}^{\infty} \frac{1}{n^2+1}$

c. $\sum_{n=2}^{\infty} \frac{1}{2n+1}$

d. $\sum_{n=2}^{\infty} n^{-n} n!$

45 [Challenging] • Target: Antiderivatives of composite expressions • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Consider the function $f(x) = x^2$. What is the value of the definite integral $\int_0^3 f(x)dx$?

a. 9

b. 12

c. 27

d. 3

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